Orchestra Warm-Up Routine

Week 7 Practice - Problem 3 (Intermediate)

The hierarchy is Instrument -> StringInstrument -> Violin. Each subclass calls super() in its constructor and layers its detail on the prior play() result.

Java solution

```
abstract class Instrument {
    // Concrete base method is needed for super.play() to compile.
    public String play() { return ""; }
}
class StringInstrument extends Instrument {
    StringInstrument() { super(); }
    @Override public String play() {
        return super.play() + "Strumming the strings";
    }
}
class Violin extends StringInstrument {
    Violin() { super(); }
    @Override public String play() {
        return super.play() + ", with a bow drawn across four strings";
    }
}
public class Main {
    public static void main(String[] args) {
        StringInstrument s = new StringInstrument();
        Violin v = new Violin();
        System.out.println(s.play());
        System.out.println(v.play());
    }
}
```

Sample output

```
Strumming the strings
Strumming the strings, with a bow drawn across four strings
```

Note: Source conflict: Instrument.play() is requested as abstract, but StringInstrument is also required to call super.play(). Java cannot call an abstract method with super. This compilable version keeps Instrument abstract and makes play() a concrete empty base method.